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Original Article

Improved estimation of the prevalence of bovine cysticercosis and the diagnostic test characteristics in the absence of a reference standard using Bayesian Latent Class models, the example of Jimma and Ambo abattoirs, Ethiopia

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ABSTRACT

Bovine cysticercosis, an infection in cattle caused by the zoonotic tapeworm *Taenia saginata*, leads to considerable global economic losses. The reported prevalence of bovine cysticercosis in Ethiopia has, until now, been underestimated due to the low sensitivity of meat inspection. This study aimed to estimate the true prevalence of bovine cysticercosis in two Ethiopian abattoirs, and evaluate the diagnostic performance of routine meat inspection, predilection-site dissection and Antigen-ELISA. A cross-sectional study was conducted and a Bayesian Latent Class Model was used to estimate the true prevalence and the diagnostic performance of the tests. The apparent prevalence of bovine cysticercosis was 2 %, 15 % and 14 % using meat inspection, predilection site dissection and Ag-ELISA, respectively. The estimated true prevalence was 39.0 % (95 %CI: 16.3, 93.7 %) in Jimma, and 31.7 % (95 %CI, 12.1, 77.9 %) in Ambo abattoirs. Predilection site dissection had moderate sensitivity (50.4 %, CI, 19.7, 96.4 %) and while Ag-ELISA had a lower sensitivity (17.3 %, CI, 5.0, 37.3 %). The sensitivity of Ag-ELISA for the detection of viable cysticerci only was 46.7 % (CI, 4.1, 91.1 %). Meat inspection had a much lower sensitivity (3.0 %, 95 %CI, 0.3–10.0 % in Jimma and 7.8 %, 95 %CI, 0.90, 21.8 % in Ambo abattoirs). Predilection site dissection has high specificity (98.9 %, CI, 96.3, 100.0), and Ag-ELISA has moderate specificity (86.9 %, CI, 77.6, 100). In this study, imperfect diagnostic tests were combined within a Bayesian framework to estimate true prevalence without relying on a perfect reference test. The findings highlight that the true prevalence of bovine cysticercosis is substantially higher than the apparent prevalence of the separate methods and that the sensitivity of current meat inspection practices is low. Therefore, adopting more sensitive diagnostics, routine incisions of the tongue and heart, and strengthened surveillance and training are suggested to improve detection and protect public health.

1. Introduction

Bovine cysticercosis (BCC) is an infection of cattle with the *Taenia saginata* metacestode larval stage (cysticercus). The *T. saginata* life cycle involves humans as final and cattle as intermediate hosts. People are infected by the consumption of raw or inadequately cooked beef containing viable cysticerci. In the intestines, the cysticerci develop into a tapeworm, leading mostly to no or mild symptoms (Allan et al., 1991). Cattle become infected principally by ingestion of feed or water contaminated with eggs from the stool of tapeworm carriers. The cysticerci develop primarily in the cattle muscles and become infective to humans after 10 weeks. After around nine months, most cysticerci degenerate and calcify, but some remain viable. BCC is generally asymptomatic; nevertheless, a great economic loss is incurred from the lowering the value of meat and condemnation of infected carcasses and meat inspection resources (Collins and Huey, 2009; Jansen et al.,

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2018b).

Taenia saginata taeniasis and cysticercosis are distributed worldwide, occurring both in the industrialized world and the global South. However, there is a higher occurrence in Latin America, some parts of Asia and Africa, particularly in Central and East African countries. In several countries, notable meat inspection–based prevalence estimates of bovine cysticercosis have been reported. These include a 7.8 % pooled prevalence in Ethiopia (Jorga et al., 2020), 5.4 % in Kenya (Oduori et al., 2024), 3 % in Rwanda (Nzeyimana et al., 2015), 3.4 % in Southern Brazil (Rossi et al., 2020), 3 % in Pakistan and 1.96 % in Indonesia (Eichenberger et al., 2020). This high prevalence is attributed to culinary habits, poor sanitary infrastructure, and lack of MI (Abunna et al., 2008; Allan et al., 2005; Regassa et al., 2009).

Meat inspection is the primary method to interrupt the transmission of *T. saginata* from cattle to humans. According to the Ethiopian Meat Inspection Guideline (MOA, 1972), carcasses found free of cysticerci during meat inspection are approved for human consumption. Carcasses with mild infections (1–5 cysticerci) need to undergo freezing treatment at −10 °C for at least ten days to kill the cysticerci before being released for consumption. Moderately infected carcasses (6–20 cysticerci) are conditionally approved, requiring mandatory cooking at high temperatures (above 56 °C). Carcasses with heavy infections (>20 cysticerci) are deemed unfit and condemned. However, the challenges in meat inspection practices mean that some carcasses may enter the food chain without proper inspection or due to the low sensitivity of current inspection methods may result in cysticerci, allowing infected meat to be sold for human consumption.

Recent abattoir surveys in Ethiopia vary widely from 2.8 % to 27.3 % (Abera et al., 2022; Fesseha and Asefa, 2023; Mohamed et al., 2021). Reported prevalence is highly variable between studies and regions. According to (Jorga et al., 2020), MI is not performed according to the Ethiopian MI standard, which may have led to an underestimation of the BCC prevalence. Nevertheless, even when performed correctly, routine MI has a known low sensitivity (<15 %) (Dorny et al., 2000), which could even fall below 1 % in populations with a low level of infection, such as cattle in Belgium (Jansen et al., 2017). Hence, the use of MI alone could underestimate the actual prevalence in cattle, implying the need for alternative detection techniques for a better estimation of the prevalence. Multiple incisions in the heart muscle resulted in a doubling of the number of infected carcasses in a Swiss study (Eichenberger et al., 2011), though this could not be confirmed in a later study in Belgian cattle (Jansen et al., 2017). Improved detection of cysticerci was reported using complete dissection of predilection sites (PS, i.e., heart, tongue, masseter muscles, esophagus, and diaphragm) in Belgian cattle (Jansen et al., 2017). On the other hand, serological techniques such as Enzyme-Linked Immunosorbent Assays detecting circulating excretory-secretory cysticercal antigens (Ag-ELISAs) (Brandt et al., 1992), and Ab-ELISA detecting specific antibodies using different types of coating antigen (Eichenberger et al., 2013) have improved the diagnosis of BCC. The Ag-ELISA detects viable cysticerci, while the antibody detection by Ab-ELISA does not differentiate between recent infections with viable cysticerci and old infections with degenerated cysticerci or past infections (Jansen et al., 2017). The sensitivity of Ag-ELISA was estimated at 92 % in cattle infected with more than 50 cysticerci and the specificity at 98.7 % (Van Kerckhoven et al., 1998; Jansen et al., 2018a).

The specificity of MI or dissection based on visual recognition of lesions could be adequate for viable cysticerci, while for degenerated or calcified lesions, molecular confirmation of cysticerci would be beneficial, especially in Ethiopia, where multiple parasitic (including other taeniid infections) or other infections could cause similar lesions. Molecular techniques such as Multiplex Polymerase Chain Reaction (mPCR) have been used for the identification of cysticerci to *Taenia* spp. (Geysen et al., 2007; Yamasaki et al., 2004). The sensitivity and specificity of *Taenia* species Multiplex PCR for confirming *T. saginata* cysticerci was estimated as 98 % and 100 %, respectively. However, the sensitivity of the Multiplex PCR could be affected by the availability of the *Taenia* DNA, especially in degenerated and calcified cysts (Jansen et al., 2017).

In Ethiopia, previous studies solely relied on routine MI to estimate the prevalence and in one case, an indirect hemagglutination serology test estimated a prevalence of 25.6 % (Kebede et al., 2008). The current study is among the first in the country to utilize multiple tests, including the serological (Ag-ELISA) test and PS dissection to estimate the true prevalence of bovine cysticercosis using a Bayesian latent class model (BLCM). Since currently available diagnostic tests are imperfect and the true status of BCC in the studied carcasses is not known, the use of BLCM combining the results of multiple diagnostic tests is appropriate. Bayesian models have been successfully implemented to estimate the prevalence and diagnostic test characteristics of BCC in cattle carcasses slaughtered in different abattoirs in Europe in the absence of a gold standard test (Eichenberger et al., 2013; Jansen et al., 2018a). Therefore, the objectives of the present study were to estimate the true prevalence of BCC and evaluate the diagnostic performance of routine MI, enhanced predilection-site dissection and Ag-ELISA using a BLCM in Jimma and Ambo abattoirs of the Oromia region, Ethiopia.

2. Materials and methods

2.1. Study area

The study was conducted in Jimma and Ambo municipal abattoirs of the Oromia region, Ethiopia. Geographically, Jimma town is located in Jimma zone 350 km South West of Addis Ababa, whereas Ambo town is the administrative center of the West Shoa zone, located 114 km west of Addis Ababa (Fig. 1). The climate of Jimma area is tropical humid, characterized by heavy rainfall ranging from 1200 to 2000 mm, and the mean annual minimum and maximum temperature range between 14.4 and 26.7 °C (NMSA, 2016). The mean annual rainfall of Ambo area ranges from 500 to 1600 mm and the maximum and minimum temperature ranges from 10 to 28 °C (NMSA, 2016). The cattle populations of Jimma and West Shoa zones were 2,560,207 and 2,294,593, respectively (CSA, 2021). These are the two zones holding large cattle populations in Oromia region, where cattle are mostly managed under an extensive system. Jimma municipal abattoir receives cattle for slaughter from Jimma zone and Southwestern Ethiopia region and has the capacity to slaughter 40–70 cattle, 10–25 sheep, and 5–10 goats per day. Ambo municipal abattoir receives cattle from the West Shewa zone and nearby East Wellega and Horo Guduru zones of western Ethiopia and has a daily slaughtering capacity of 20–40 cattle. Both abattoirs are under the governance of their respective municipality and are permitted to provide slaughter services of cattle and some small ruminants and supply the meat for consumption and sale to restaurants and butchers. In Ethiopian municipal abattoirs, cattle are presented for slaughter at the slaughterhouse by the butcher men, afterwards the meat is sold for human consumption in butcher shops or restaurants.

Meat inspection in the two abattoirs is performed in such a way that ante mortem inspection is carried out on a routine basis, while post-mortem inspection is performed irregularly. At Jimma abattoir during post-mortem inspections, more emphasis is given to the liver and lung for cystic echinococcosis and the carcass lymph nodes for tuberculosis lesions. Visual inspection of the carcass is usually conducted for BCC, and incisions to the triceps muscle are performed whenever the carcass is suspected of cysticerci during visual inspection of carcass surfaces. In contrast, at Ambo abattoir, post-mortem inspections are rarely performed, and irregularly for BCC, depending on the presence of meat inspectors in the abattoir during slaughter.

2.2. Study subjects

Cattle of any sex, age, and breed brought to the two municipal abattoirs by butcher men for slaughter for the purpose of human consumption during the study period were eligible for inclusion in the

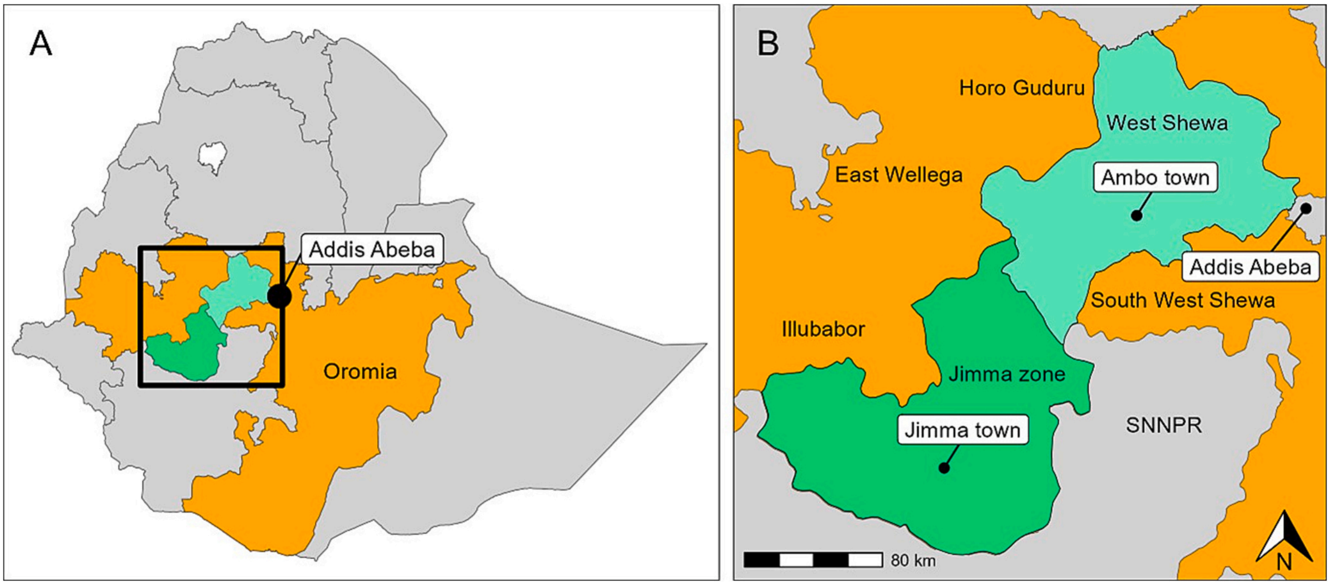

Fig. 1. A) Map of the study area in Oromia region of Ethiopia. B) The abattoirs are located in Jimma zone and West Shewa zone of Oromia region. Note: SNNPR: Southern Nations Nationalities and peoples' region.

study.

2.3. Sample size

The sample size for estimating the prevalence of BCC based on the dissection of PS and serology was calculated using Thrusfield's formula (Thrusfield, 2005). Considering a 95 % confidence level, 5 % precision, and 50 % expected prevalence, the calculated sample size was 384 animals from each study abattoir, with a total number of 786 animals. A slightly more number of animals were sampled (809) and the final sample size analyzed was 577.

2.4. Study design

A cross-sectional abattoir-based investigation of BCC was conducted on cattle carcasses using results from MI at each abattoir, if performed, dissection of PS and serology (Ag-ELISA) as diagnostic techniques. Regular abattoir visits were organized two days per week during non-fasting seasons of the years between July 2018 and July 2020. On each visit, about six to ten carcasses were sampled based on the number of animals presented for slaughter and owners' willingness to sell the PS. When more than ten carcasses are available, carcasses were selected conveniently at irregular intervals, which is partially non-random sampling approach. This study was carried out on cattle carcasses slaughtered in the abattoir for the purpose of human consumption; hence, live animals were not used in the study.

2.5. Postmortem inspection and sample collection

The selected cattle were given an identification number, which was transferred to the successive samples during the abattoir and laboratory work. Data regarding, sex, age, body condition, and breed of the slaughtered cattle were recorded. Blood samples from the selected animals were collected in a 10 ml plain blood collection tube immediately after cutting the throat at the beginning of the slaughter process. Subsequently, each selected carcass's heart, tongue, masseters, diaphragm, and esophagus were collected and separately marked in a plastic bag for complete dissection. The viscera (liver, lung, omentum, peritoneum, and mesentery) were carefully inspected for *Echinococcus* spp. and *Taenia hydatigena* cysts. This was done as the Ag-ELISA is not *T. saginata* species-specific (Dorny et al., 2004). The collected samples were labeled and transported to the laboratory. Reports of routine MI regarding the BCC status of the carcass based on visual inspection of the carcass surfaces and incision of triceps muscle were collected from meat inspectors in charge, whenever MI was conducted.

2.6. Dissection of predilection sites (PS)

Dissection was performed by slicing the PS in 0.5 cm thick slices by the researchers. All surfaces were thoroughly checked for the presence of cysticerci. The number and stage of the cysticerci were recorded at the organ level. Cysticerci were recorded as viable, degenerated, or calcified by visual inspection and palpation as described by Ogunremi and colleagues (Ogunremi et al., 2004). Cysticerci were classified as viable if soft to touch and fluid-filled, with or without a visible scolex; as degenerated if hard to touch or creamy, greenish, or yellowish discolored; and calcified when the content was hard creamy, or solid and perceptible when sliced.

2.7. Enzyme-linked immunosorbent assay for the detection of circulating antigens (Ag-ELISA)

Blood samples were kept overnight at 4 °C, centrifuged and serum was collected in Eppendorf tubes and stored at −20 °C in Ambo University veterinary laboratory until analysis. The Ag-ELISA was performed in the National Animal Health Diagnostic and Investigation Center (NAHDIC) serology laboratory, Sebeta, Ethiopia.

Circulating cysticercus antigens were detected using monoclonal antibodies, in an antigen-trapping ELISA in duplicate. The Ag-ELISA was conducted using a commercial kit (apDia, Belgium). Pre-treatment of samples and controls was done with trichloroacetic acid and neutralization buffer as per the recommended procedure. Then, pretreated samples/controls were added into monoclonal antibody-coated wells, followed by incubation to facilitate the binding of antigen in the serum with the coated antibody. Unbound proteins were removed by washing and the antigen-antibody complex in the wells was detected by peroxidase conjugate. The unbound conjugate was removed by washing, followed by incubation with chromogen solution. Finally, a stopping solution was used to end the enzymatic reaction and the optical density (OD) was read at 450 nm with a reference wavelength of 600-650 nm in a microplate reader. As per the manufacturer's guide, the Ag-ELISA result was considered valid when the OD value for the positive control is above 1.00 and that of the negative control is below 0.1, otherwise, it was considered invalid. All invalid results were removed from the analyses.

The calculation of the Ag-ELISA cut-off was based on the change-point analysis method (Lardeux et al., 2016) using R version 4.0.2 (R Core Team, 2020). Using the change-point package (Killick and Eckley, 2014), the Pruned Exact Linear Time (PELT) algorithm was selected to detect the change point. For this purpose, each of the OD values was changed to S/P ratio (sample to positive ratio) using the formula S/P = (TS –NC) / (PC –NC), using the mean OD value of TS (test sample), NC (negative control), and PC (positive control). According to the change point analysis, a cut-off S/*P* value of 0.661 was obtained and used to discriminate the Ag-ELISA result as positive or negative.

2.8. Tissue sample collection and DNA extraction

Up to four suspected cysticerci per positive carcass were collected during PS dissection and stored in 1.5 ml tubes containing 70 % ethanol. Each cyst was collected separately in a labeled tube using new or properly disinfected utensils. Until DNA extraction was performed, all cysts/lesions were kept in the Ambo University veterinary laboratory. DNA extraction was done using DNeasy Blood and Tissue kit (QIAGEN, Hilden, Germany), according to the manufacturer's instruction at NAHDIC, Ethiopia.

2.9. Molecular analysis

Multiplex PCR for *Taenia* spp. was conducted according to Yamasaki and colleagues (Yamasaki et al., 2004). The mPCR reaction is designed to amplify the cytochrome *c* oxidase subunit 1 gene (*cox*1), resulting in a unique amplicon length for *T. saginata* (827 bp), *T. asiatica* (269 bp), and American/African genotypes of *T. solium* (720 bp) and Asian genotypes of *T. solium* (984 bp). Two sets of primers were used, and the forward primer was specific for each species while the reverse primer was common to all species. The mPCR reaction was carried out in a final reaction volume of 25 μl containing 12.5 μl PCR Master Mix (2×), 5 μl RNase-Free Water, 2.5 μl Primer mix (containing 2 μM of each forward, and 4 μM of the reverse primer), and 5 μl template DNA. DNA fragments were amplified according to the following thermal cycling conditions: 95 °C/15 min for initial denaturation; 30 cycles of 94 °C/30s for denaturation, 58 °C/90s for annealing, 72 °C/90s for extension; 72 °C/ 10 min for the final extension. Amplified PCR products were visualized by electrophoresis in 2 % agarose gel that were stained with ethidium bromide and viewed under UV light.

Taenia spp. mPCR-negative reactions were tested as per Trachsel and colleagues (Trachsel et al., 2007) using a mPCR procedure with specific primers for *E. multilocularis* (395 bp), *E. granulosus* (117 bp), and *Taenia* spp. (267 bp) targeting the *nad1*, *rrnS,* and *rrnS* genes, respectively. The final mPCR reaction volume of 25 μl was run at a thermal condition involving initial activation at 95 °C for 15 min, followed by 94 °C for 30 s, 58 °C for 30 s, 72 °C for 10 s, and 40 cycles.

Lesions that were negative to both mPCR procedures mentioned above were analyzed using mPCR for *Sarcocystis* spp. following the procedure followed by Rubiola and colleagues (Rubiola et al., 2020) using specific primers for *S. cruzi* (300 bp), *S. hirsute* (100 bp), *S. hominis* (420 bp), *S. bovifelis* (700 bp), and *Sarcocystis* spp. (200–250 bp) targeting the 18S rRNA and mtDNA COI genes. The final mPCR reaction volume of 25 μl was used and the thermal cycling conditions involved denaturation at 95 °C for 3 min, followed by 35 cycles at 95 °C for 1 min, 58 °C for 1 min, and 72 °C for 30 s, and final extension 72 °C for 3 min. Molecular analysis was done in Belgium, Gent University, Faculty of Veterinary Medicine, Laboratory of foodborne parasitic zoonoses.

2.9.1. Sequencing

Thirteen lesions showing a doubtful or light band during *Taenia* spp. mPCR or those showing a *Taenia* spp. band during *Echinococcus* spp. mPCR were sent to Eurofins Genomics (Luxembourg) for sequencing based on the *rrnS* genes. The sequences obtained were compared to available sequences from the GenBank database using the Basic Local Alignment Search Tool (BLAST).

2.10. Data management and analysis

Data from the abattoir (MI, animal data), dissection of PS, mPCR, sequencing, and Ag-ELISA were compiled in a Microsoft Excel spreadsheet. Descriptive analysis was performed using R version 4.0.2 (R Core Team, 2020) and data were summarized and presented using frequency tables, percentages, and figures. For the carcasses for which half-dissection was made for the heart, tongue, and masseter muscle (n = 82), while the diaphragm and esophagus were dissected fully, the numbers of cysticerci detected from the half-dissected heart, tongue, and diaphragm (n = 18) were doubled to correct for the half-dissection. For dissection results, carcasses from which multiple suspected cysticerci tested were considered *T. saginata* positive if at least one of the suspected lesions was confirmed molecularly. The association of BCC with age, sex, and breed of slaughtered cattle was not computed, as almost all slaughtered cattle in our sample were adult males of local breeds.

To estimate the sensitivity and specificity of dissection and Ag-ELISA, along with the true prevalence of BCC in cattle slaughtered in each of the abattoirs, data were analyzed using BLCM. In each abattoir, the number of carcasses for each possible test result combination for dissection and Ag-ELISA (positive or negative) were modeled using a multinomial distribution, assuming that the sensitivity and specificity of the tests do not vary in both abattoirs. Dissection detects visible cysticerci (all stages) and Ag ELISA detects circulating Ag (primarily viable stages). Firstly, analyses were conducted based on all stages, so the latent class was infection with BCC (any stage of cysticerci). Since only viable cysticerci are a public health risk, the analysis was also repeated using data from viable cysticerci of PS dissection, in which case the latent class was BCC infection with viable cysticerci. Minimally informative priors were used for the sensitivity of dissection, the sensitivity and specificity of Ag-ELISA, and the prevalence in both abattoirs (Beta (1,1) (mean = 0.50, SD ≈ 0.29). As the specificity of dissection was expected to be very high (because only carcasses with molecularly confirmed lesions were considered PS dissection positive for the BLCM analysis), a restrictive prior was used for this test. The prior for dissection specificity, Beta(106.7, 1.408) with mean ≈ 0.99, SD ≈ 0.01, was generated using the R package 'PriorGen' (Pateras and Kostoulas, 2023) using a median estimate of 99 % and a lower limit of the 99 % CI of 95 %. We acknowledge that the specificity estimates are thus strongly influenced by the prior distributions, offering limited additional information beyond the assumptions; however, the use of informative priors was necessary to ensure model identifiability and convergence. The models, generated using the function "template_huiwalter", were fit to the data using the runjags package (Denwood, 2016), running two chains of 10,000 iterations each with thinning (interval of 5) after an initial 5000 burn-in period. The sensitivity of PS dissection and Ag-ELISA were allowed to be correlated, as the infection intensity may influence the likelihood of detection by both PS dissection and Ag-ELISA, relaxing the usual assumption of conditional independence. The parameter estimates were summarized using the median and 95 % credibility intervals. Model convergence was assessed via trace and autocorrelation plots, and the potential scale reduction factor (PSRF). The effective sample size was ensured to be over 1000 independent iterations for each of the parameters. The sensitivity and specificity of meat inspection in both abattoirs were derived from posterior positive probability (PPP) values, as described by Olsen and colleagues (Olsen et al., 2022). Visual inspection of trace plots showed good chain mixing and no trends, all parameters achieved an effective sample size greater than 1000, and the potential scale reduction factor (Gelman–Rubin statistic) for every parameter was ≤1.01, indicating satisfactory convergence.

3. Results

As described in Fig. 2 and supplementary file 1, 809 carcasses were included in this study; however, complete data for PS dissection and Ag-ELISA were obtained from 577 carcasses (329 from Ambo, and 248 from Jimma). For the remaining 232 carcasses, data were incomplete either for PS dissection ($n = 96$, not all PS could be purchased) and molecular confirmation of the collected cysticerci (n = 9, sample not available) or Ag-ELISA ($n = 127$, no valid result). From here onwards we continue with the 577 complete set of samples for analysis and BLCMs.

Out of 473 carcasses (104 carcasses had no MI result for echinococcosis out of 577 carcasses) that could be checked for *Echinococcus* spp., 39 (8 %) were carrying various numbers of *Echinococcus* cysts in the liver or lungs. No *T. hydatigena* was observed. However, the detection of *Echinococcus* cysts was not significantly associated with Ag-ELISA positivity.

Meat inspection detected cysticerci in 8 carcasses (2.0 % overall), comprising 3 cases in Jimma (1.2 %) and 5 cases in Ambo (2.6 %) out of the 443 carcasses with available MI results (134 carcasses had no MI result for BCC out of 577 carcasses). Predilection site dissection followed by molecular confirmation identified 90 out of 577 carcasses, resulting in an apparent prevalence of 15.6 %. Antigens were detected using Ag-ELISA in 81 out of 577 carcasses (14 %). Of 485 carcasses negative to both MI and PS dissection, 61 (13 %) were positive using Ag-ELISA. Full agreement was observed for 424 negative carcasses by all the diagnostic tests.

3.1. Dissection of PS

About 18 % of 577 carcasses had a suspected cysticerci lesion in at least one PS during the dissection of PS. The suspected cysticercus lesions were found mostly in the tongue (11 %) and the heart (8 %), followed by the masseter muscle (3 %), diaphragm (1 %), and esophagus (1 %). Out of the 260 suspected cysticercus lesions observed during the dissection of the PS, 130 (50 %) were degenerated, 76 (29 %) were viable and 54 (21 %) calcified. Most of the viable lesions (40/76) were

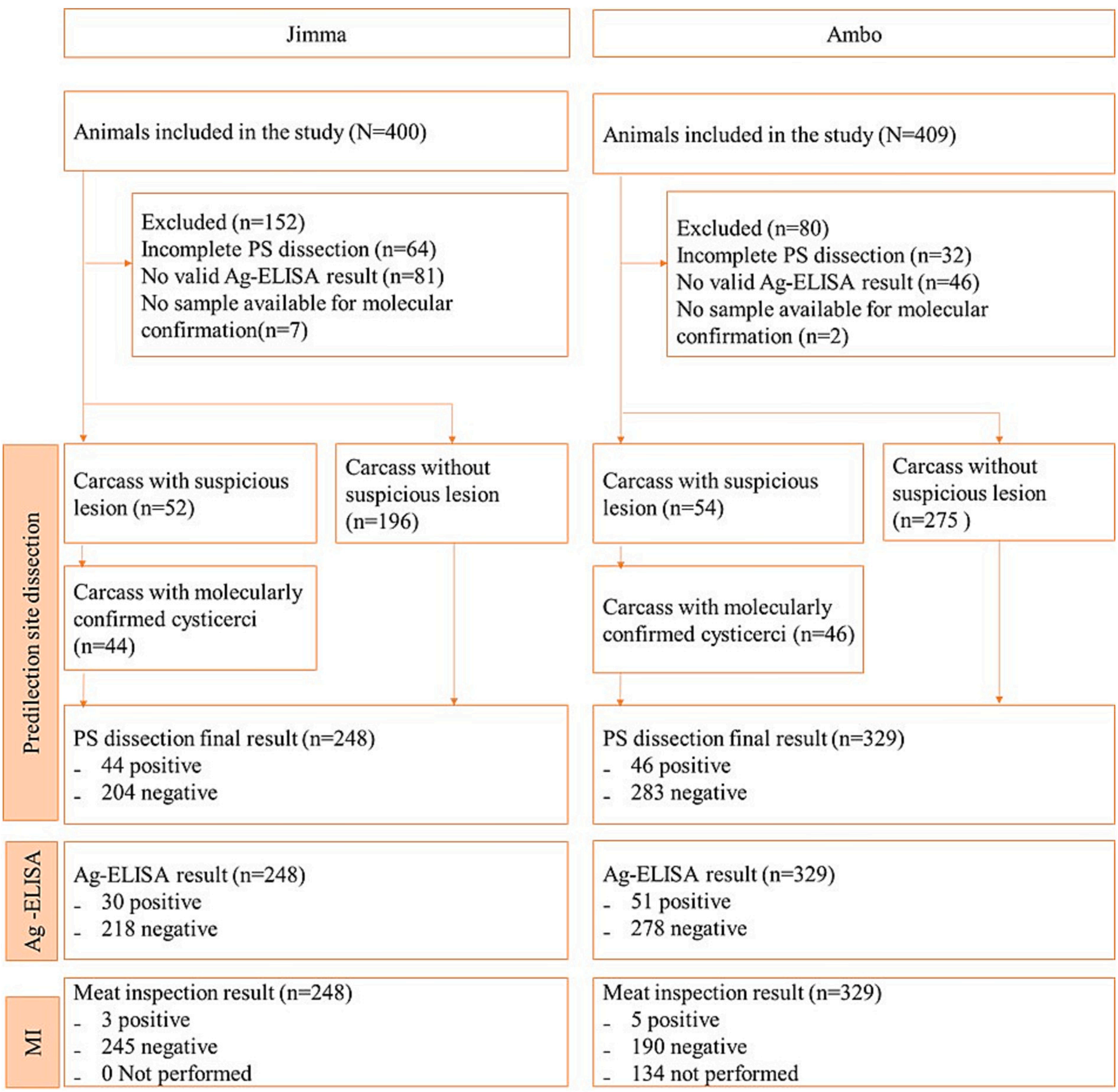

Fig. 2. Flow diagram according to STARD (Standards for the reporting of diagnostic accuracy) for PS dissection, antigen- enzyme-linked immunosorbent assay and meat inspection for the diagnosis of bovine cysticercosis.
Note: MI: meat inspection, PS: predilection site, Ag-ELISA: Antigen- Enzyme-Linked Immunosorbent Assay, N: Number of samples.

found in the tongue, while most of the calcified lesions (43/54) were found in the heart (Table 1).

As shown in Table 2, suspected cysticerci lesions were detected in a single PS in 90 carcasses. More than half of these detections were in the tongue (48 carcasses). Detection from multiple PS was observed in 16 carcasses. In 54 carcasses, only one suspected lesion was found. In 13 carcasses, six or more lesions were observed with a maximum of 14 per carcass (Fig. 3).

3.2. Molecular analysis

Taenia spp. mPCR confirmed 101 lesions as *T. saginata* cysticerci from 133 suspected lesions analyzed (at least one cyst-like lesion per suspected lesion carrying carcass was tested). Thirty-two *Taenia* spp. mPCR-negative reactions were analyzed using *Echinococcus* mPCR. Two of them were confirmed as *E. granulosus*, of which one was a viable cyst collected from the heart and the other one was calcified, which was collected from the diaphragm. Fifteen lesions showed a *Taenia* spp. band that was not identified by *Taenia* spp. mPCR. Sequencing of those lesions confirmed a further 9 as *T. saginata* out of 13 analyzed (for two lesions insufficient sample was available for sequencing). Seventeen lesions negative or with doubtful PCR results were checked for *Sarcocystis* spp. using *Sarcocystis* mPCR and four of them were confirmed as *Sarcocystis* spp.

Hence, molecular analysis confirmed 110 (83 %) suspected cysticerci as *T. saginata*. Based on the stage, 36 (95 %) out of 38 viable, 57 (76 %) out of 75 degenerated and 17 (85 %) out of 20 calcified lesions were confirmed as *T. saginata* cysticerci. At carcass level, 90 (85 %) out of 106 carcasses carrying suspected lesions were confirmed to be positive for *T. saginata* cysticerci (BCC). Lesions from two carcasses were confirmed as *E. granulosus* and those from four carcasses were *Sarcocystis cruzi*. Lesions from four carcasses showing a *Taenia* spp. band during *Echinococcosis* mPCR were not confirmed by sequencing and were considered negative. Suspected lesions from six carcasses, all of which were degenerated, were negative using all the PCR procedures.

3.3. Ag-ELISA

Of all 577 tested carcasses, 81 (14 %) tested positive using the Ag-ELISA. Nineteen of the Ag-ELISA-positive carcasses were also carrying molecularly confirmed cysticerci, 10 of which were carrying viable cysticerci in the PS. Notably, 62 carcasses had a positive Ag-ELISA despite no cysts being found by either routine MI or PS dissection, most of which could represent light infections undetectable by lesion inspection, while others might be false positives due to antigen cross-reaction of other parasites. Of the 39 carcasses carrying *Echinococcus* cysts either in the liver, lung, or in both, seven were Ag-ELISA positive, three of which were also carrying molecularly confirmed cysticerci.

3.4. Diagnostic accuracy of the different tests and prevalence estimation

The diagnostic test results obtained for the detection of BCC using PS dissection, MI and Ag-ELISA at the two abattoirs were given below in Table 3.

Table 1
Distribution and stage of suspected cysticerci during PS dissection (n = 577 carcasses).

Predilection site (PS)	No. positive PS (%)	No. of lesions detected	Stage of suspected cysticerci lesions		
			Viable (%)	Degenerated (%)	Calcified (%)
Masseter	16 (3)	26	6 (23)	15 (58)	5 (19)
Tongue	63 (11)	126	40 (32)	83 (66)	3 (2)
Esophagus	6 (1)	6	2 (33)	2 (33)	2 (33)
Heart	46 (8)	94	23 (25)	28 (30)	43 (46)
Diaphragm	7 (1)	8	5 (63)	2 (25)	1 (13)
Total	138 (5)	260	76 (29)	130 (50)	54 (21)

Table 2
Distribution of suspected cysticerci lesions across the PS in the examined carcasses (n = 577). "Positive" indicates the presence of ≥1 cysticercus in that organ; "Negative" indicates none detected.

Tongue	Heart	Masseter	Diaphragm	Esophagus	Number of carcasses
Negative	Negative	Negative	Negative	Negative	471
Negative	Negative	Negative	Negative	**Positive**	1
Negative	Negative	Negative	**Positive**	Negative	2
Negative	Negative	**Positive**	Negative	Negative	8
Negative	**Positive**	Negative	Negative	Negative	31
Positive	Negative	Negative	Negative	Negative	48
Negative	**Positive**	Negative	**Positive**	Negative	1
Positive	Negative	**Positive**	Negative	Negative	1
Positive	**Positive**	Negative	Negative	Negative	6
Positive	**Positive**	Negative	**Positive**	**Positive**	1
Positive	**Positive**	**Positive**	Negative	Negative	2
Positive	**Positive**	**Positive**	Negative	**Positive**	2
Positive	**Positive**	**Positive**	**Positive**	Negative	1
Positive	**Positive**	**Positive**	**Positive**	**Positive**	2

The estimates from the BLCMs for the sensitivity and specificity of PS dissection and the Ag-ELISA as well as the prevalence for any stage of cysticerci and viable cysticerci in both abattoirs are separately given in Table 4. The sensitivity of PS dissection for any stage of cysticerci was estimated at 50.4 % (19.7–96.4 %), whereas it was 23.4 % (2.8–83.0 %) for viable cysticerci only. The sensitivity of the Ag-ELISA for the detection of any stage of cysticerci was 17.3 % (5.0–37.3 %), whereas it was 46.7 % (4.1–91.1 %) for the detection of viable cysticerci only. The specificity of dissection of PS for any stage of cysticerci was estimated at 98.9 (95 % CI 96.3, 100.0), and 86.9 % (67.4–99.9 %) for Ag-ELISA, whereas it was 98.7 % (96.8–100 %) for dissection and 89.0 % (77.6–100 %) for Ag-ELISA for viable cysticerci only. The model estimated a true prevalence of BCC [any stage] of ~39 % in Jimma and ~32 % in Ambo, although the 95 % CrI were broad [Jimma: 16.3–93.7 %; Ambo: 12.1–77.9 %], reflecting considerable uncertainty. For viable cysticerci only, the median prevalence was around 12–13 % for both abattoirs, but with intervals spanning from <1 % up to ~60–80 %, indicating low precision in those estimates.

The sensitivity and specificity of MI in both abattoirs were estimated based on the final model and are shown in Table 5. The sensitivity of meat inspection in Jimma abattoir was 3.0 % (0.3, 10.0 %) and in Ambo 7.8 % (0.9, 21.8 %) for carcasses for which MI was performed. When considering the carcasses that were not inspected in Ambo (n = 134) as MI negative, the estimated sensitivity of MI in Ambo decreased to 4.0 % (95 % CI 0.6–11.7 %). The specificity of meat inspection in both abattoirs was high.

4. Discussion

Bovine cysticercosis has a worldwide occurrence, and routine MI is the method used for its diagnosis and control. However, the low sensitivity of MI makes this method unreliable to estimate the actual prevalence of BCC (Dorny et al., 2000). The MI in the present study abattoirs was mostly done by making one deep incision into the triceps muscle of both sides. Hence, the sensitivity of MI, if performed, was low, being 3.0 % (0.3–10 %) at Jimma and 7.8 % (0.9–21.8 %) at Ambo abattoir. When the non-inspected carcasses are included, the sensitivity of MI at Ambo decreased to 4.0 % [0.6–11.7 %], indicating that the sensitivity of meat inspection in both abattoirs is low. MI is not performed regularly at Ambo abattoir, partly due to the lack of regular transportation services for meat inspectors, which is exacerbated by security issues and the timing of slaughter (during the night), which restricts their access to alternative transport systems. The Ethiopian MI Regulation (MoA, 1972) dictates the visual inspection and incision of the heart, external and

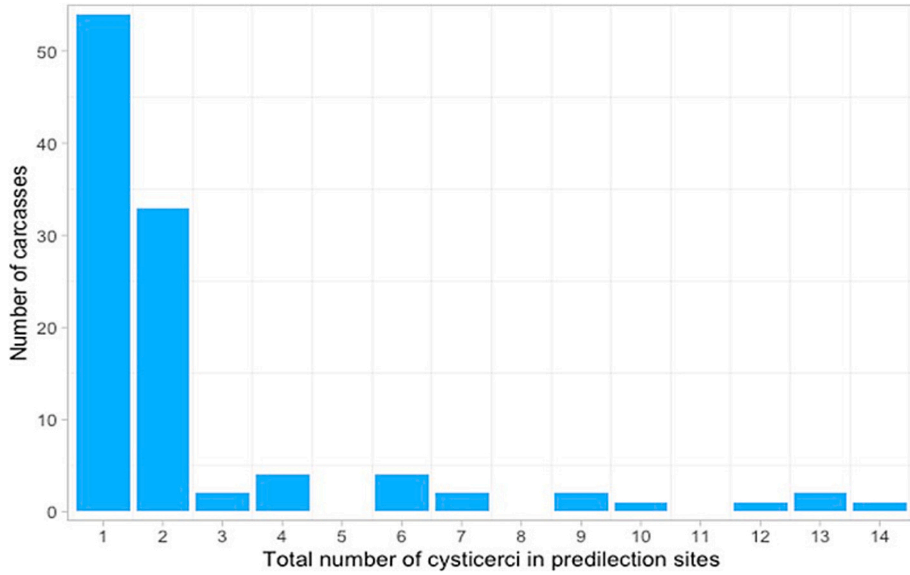

Fig. 3. Number of suspected cysticerci in predilection sites in carcasses carrying cysticerci (n = 106).

Table 3
Test combination results of meat inspection, dissection of predilection sites and Ag ELISA in two abattoirs in Ethiopia (for any stage of cysticerci).

Abattoir	Meat inspection results	PS dissection results	Ag-ELISA results	Number of carcasses
Jimma	0	0	0	176
Jimma	0	1	0	40
Jimma	0	0	1	20
Jimma	0	1	1	9
Jimma	1	1	0	2
Jimma	1	1	1	1
Ambo	0	0	0	138
Ambo	0	0	1	30
Ambo	0	1	0	19
Ambo	0	1	1	3
Ambo	1	1	1	3
Ambo	1	0	1	1
Ambo	1	1	0	1
Ambo	NP	0	0	96
Ambo	NP	1	0	24
Ambo	NP	0	1	10
Ambo	NP	1	1	4

Note: 0 = negative test result, 1 = positive test result, PS = predilection site, NP = not performed, Ag-ELISA = antigen ELISA.

Table 4
Diagnostic accuracy and prevalence of BCC (any stage of cysticerci and viable cysticerci only) in cattle of Jimma and Ambo abattoirs as estimated using BLCMs.

	BCC (any stage)	BCC (viable only)
	Median (95 % CI)	Median (95 % CI)
Sensitivity dissection	50.4 (19.7, 96.4)	23.4 (2.8, 83.0)
Sensitivity Ag-ELISA	17.3 (5.0, 37.3)	46.7 (4.1, 91.1)
Specificity dissection	98.9 (96.3100.0)	98.7 (96.8, 100)
Specificity Ag-ELISA	86.9 (67.4, 99.9)	89.0 (77.6, 100)
Prevalence of BCC at Jimma	39.0 (16.3, 93.7)	12.0 (0.9, 80.0)
Prevalence of BCC at Ambo	31.7 (12.1, 77.9)	13.1 (0.2, 63.9)

Note: BCC=Bovine cysticercosis, CI = Credible Interval.

Table 5
The sensitivity and specificity of meat inspection in cattle slaughtered at Jimma and Ambo abattoirs of Ethiopia as estimated using BLCMs.

	Median (95 % credible interval)
Sensitivity of meat inspection in Jimma	3.0 (0.3, 10.0)
Sensitivity of meat inspection in Ambo	7.8 (0.9, 21.8)
Specificity of meat inspection in Jimma	99.9 (94.4, 100)
Specificity of meat inspection in Ambo	99.6 (97.1, 100)

internal masseter muscle, tongue, triceps, thigh muscle, neck muscles, and the diaphragm; and if necessary, incision of the intercostal muscle, liver, lung, kidney, and esophagus, but these rules are not followed by either of the abattoirs. In Ethiopian abattoirs, incisions to all the recommended PS are performed primarily within the framework of research projects to estimate prevalence. Routinely, in most abattoirs incision into the triceps muscle is commonly practiced. The same is true in the present study abattoirs, an incision of the triceps muscle detected cysticerci in 2 % of the inspected carcasses, which could have been higher if all the recommended PS would have been incised. This implies the need for reinforcement on the implementation of routine MI procedures in Ethiopian abattoirs.

The true prevalence of BCC was 39.0 % (16.3–93.7 %) in Jimma abattoirs. Although the credible intervals are very wide, this value is considerably higher than previous prevalence reports based on routine MI from Jimma, ranging between 2.9 % and 5.1 % (Firew and Moges, 2014; Tolosa et al., 2009). The 31.7 % (12.1–77.9 %) prevalence of BCC in Ambo abattoir in this study is the first report in Ambo, so there are no previous reports to compare it with. Higher BCC prevalence ranging between 26.3 and 33.5 % has been reported from Hawasa (Abunna et al., 2008) and Jigjiga (Abera et al., 2022) abattoirs located in Eastern and Southern Ethiopia, respectively, all based on MI performed by incision of several recommended muscles/organs following the Ethiopian meat inspection standard. These apparent prevalence are lower than the true prevalence estimated in these study abattoirs, indicating that the true prevalence in these abattoirs could be higher than what has been reported so far. However, these results should be interpreted with caution due to the wide credible intervals surrounding the prevalence estimates, and the assumption in the BLCM that the diagnostic accuracy of both tests is identical across the slaughterhouses, an assumption that may not hold in practice. A true prevalence of BCC estimated using a similar Bayesian approach in Swiss cattle population showed a prevalence of 16.5 % (Eichenberger et al., 2013)' where the apparent prevalence by enhanced meat inspection (multiple incisions into the heart muscle) is up to 4.5 %. A similar study in Belgian cattle population estimated a true prevalence of 33.9 % (Jansen et al., 2018a), where the average apparent prevalence based on official data is 0.23 %. Though the diagnostic tests used in the different studies are not always the same, these findings depict the prevalence of BCC reported based on MI (which also varies with countries and abattoirs) so far is an underestimation of the true prevalence.

Ethiopia is a multi-cultural and diverse ethnic country with variations in the prevalence and level of infection of BCC and taeniosis related to differences in the sanitary and raw meat-eating habits in different regions. Also, the MI procedure followed in different abattoirs in the country varies with the availability of the abattoir infrastructure and the meat inspectors' capability and experience. Therefore, the present study abattoirs may not be representative of other abattoirs in the country, hence care should be taken not to generalize the current estimated prevalence and test characteristics to other regions of Ethiopia. The current estimate of the sensitivity of MI might be an overestimate as the presence of researchers during meat inspection could have influenced the presence, practice and performance of the inspectors, thereby increasing the diagnosis of BCC.

Molecular methods confirmed 95 % of the viable, 76 % of degenerated and 85 % of calcified lesions as *T. saginata*, with an overall of 83 % of the suspected cysticerci in the carcasses as *T. saginata*. The overall confirmation rate of the taenia PCR and sequencing in the present study is comparable with other reports (Geysen et al., 2007; Jansen et al., 2017). The use of *Echinococcus* spp. mPCR followed by sequencing improved the overall confirmation rate of *T. saginata* cysticerci in the present study. Two of the cysticerci that were visually classified as viable were negative by the multiplex PCR test, one of which was later confirmed as *E. granulosus.* The *E. granulosus* case was obtained from the diaphragm, from which the lungs and the liver were also infected with cystic echinococcosis, signifying the possibility of these tissues carrying small fluid-filled hydatid cyst lesions that resemble *T. saginata* cysticerci during visual inspection or it could be contamination of the sample during sample collection. Lesions that were negative to all the mPCR tests could be due to the insufficient amplifiable DNA in degenerated BCC lesions (Abuseir et al., 2006) or other lesions such as old abscesses, fat tissue, and muscle facie (Geysen et al., 2007). The detection of *Echinococcus*, *Sarcocystis* spp., unidentified *Taenia* spp., and other unspecified lesions in the PS suggests the presence of multiple parasitic or other infections. In Ethiopia, cattle are mostly managed under an extensive system, where the veterinary service is not well developed, which could allow multiple infections. Hence, this finding pointed out the necessity of laboratory confirmation of lesions visually detected during MI or dissection for research purposes.

According to the BLCM, the sensitivity of PS dissection was estimated at 50.4 % (19.7–96.4 %), which is lower compared to 69.8 % estimated in Belgian abattoirs (Jansen et al., 2018a). According to Jansen and colleagues (Jansen et al., 2017), about 76.1 % of the cysticerci are present in tissues other than the PS, and thus, all the cysticerci cannot be found during PS dissection. A sensitivity of 50.4 % shows that most of the cysticerci are found in other parts of the carcass; hence, the dissection of the whole carcass would be important to discover all the cysticerci in the bovine carcass in the Ethiopian cattle, which is not feasible for a large-scale study. During the dissection of PS, more cysticerci were revealed in the tongue, followed by the heart compared to the masseter muscle, diaphragm and esophagus, which is in line with other reports from Africa (Jorga et al., 2020; Opara et al., 2006). In contrast, lower detection in the tongue compared to the heart, masseter, and diaphragm was reported in Belgian cattle upon dissection of PS (Jansen et al., 2017), and in Brazilian cattle upon complete dissection of the whole carcass (Lopes et al., 2011). This difference suggests a distinct distribution pattern of cysticerci in infected carcasses in Ethiopian cattle, where the tongue and heart are more commonly affected and other predilection sites may be relatively less involved. This could, reflect parasite strain variation (Scandrett et al., 2009), host factors (Wanzala et al., 2003), or environmental conditions (Maeda et al., 1996). These findings reveal the need for considering the tongue and heart incisions to improve the sensitivity of routine MI in Ethiopian abattoirs.

The low sensitivity of Ag-ELISA 17.3 % (5.0–37.3 %) to any stage of cysticerci in this study could be related to the fact that the Ag-ELISA detects only viable cysticerci, not degenerated or calcified cysticerci. In this study, only 21 % (19/90) of the BCC-positive carcasses during PS dissection carried viable cysticerci and the rest were either degenerated or calcified. The model for viable cysticerci alone estimated sensitivity of Ag-ELISA to be 46.7 % (4.1–91.1 %), which is slightly higher compared to the former. Jansen et al. (2017) reported the sensitivity of Ag-ELISA to be 40 % for carcasses carrying viable cysticerci from Belgium. Van Kerckhoven and colleagues (Van Kerckhoven et al., 1998) reported a decrease of sensitivity from 92.0 % to 12.8 % in carcasses with lighter infections carrying less than 50 viable or dead cysticerci. The low number of cysticerci per carcass was unexpected in this Ethiopian assumed highly endemic area where most of the cattle-rearing communities live close to their cattle, allowing more direct transmission, which could lead to heavily infected carcasses. In addition, in endemic areas, the higher prevalence of BCC is mostly related to poor sanitary conditions and raw meat consumption habits (Kumar and Tadesse, 2011). However, the light infections in the studied abattoirs could be related to the reduction of open defecation due to the progressive health extension program that had improved access to sanitation in the last two decades. In this regard, Ethiopia registered a reduction in the proportion of the population defecating in the open from 81.9 % in 2000 to 32.9 % in 2015 (Abebe and Tucho, 2020), which could reduce the heavy pasture contamination with taeniid eggs. The light infections also suggest that infections originate from more indirect sources, such as contaminated feed and water, and contaminated pasture, and not via contact that is more direct with tapeworm carriers, where cattle would have access to proglottids or high egg concentrations. Low infection levels indicate indirect transmission and a dispersal of eggs in the environment. Therefore, considering the lower detection rate of light infections by the diagnostic methods applied, it is presumed that carcasses carrying a fewer number of cysticerci might easily be passed undetected (Dorny et al., 2002; Dorny et al., 2000).

Even though a high prevalence of BCC has been estimated, the majority of cysticerci are either degenerated or calcified, and as such are not a risk for further transmission to humans. Nevertheless, the 12.0 % (0.9–80.0 %) and 13.1 % (0.2–63.9 %) prevalence estimated for viable cysticerci in Jimma and Ambo abattoirs, respectively, is considerable. Consumption of raw or undercooked beef is a common practice in most Ethiopian communities including Jimma and Ambo. Around 89 % of Jimma and 75 % of Ambo residents consume raw meat (Jorga et al., 2020), placing the community at significant risk of *T. saginata* infection when beef contains viable cysticerci. These finding highlights the need for strengthened meat-inspection protocols, public education on safe meat consumption, and broader control measures to interrupt transmission from cattle to humans.

As a limitation, first, it was difficult to obtain complete MI data because of inefficient MI practices in the study abattoirs. Some butchers were unwilling to sell all required PS, leading to convenience sampling based on carcass availability and owner willingness, which may limit external validity, reduce generalizability to the broader cattle population and affect accurate estimation of sampling error. This sampling bias likely influenced the prevalence and test-characteristic estimates. Second, due to the local slaughter process and incomplete MI practices (e.g., inconsistent incisions in the liver/lung), it was not always possible to thoroughly check for the presence of *Echinococcus* cysts or *T. hydatigena*. Third, no perfect gold standard exists for detecting cysticerci and the lack of control for clustering by herd or origin could affect the accuracy and interpretation of test performance and prevalence estimation. Finally, our BLCM assumed equal sensitivity and specificity of PS dissection and Ag-ELISA across both abattoirs and relied on informative priors; these assumptions were necessary for identifiability but may influence the posterior estimates, particularly in populations with low infection burdens. These limitations highlight the need for future studies that incorporate larger datasets, alternative model structures, or prior distributions derived from more extensive empirical data to refine the robustness of Bayesian prevalence estimates.

5. Conclusions

The true prevalence of BCC in the present study is considerably higher than previously reported. By combining PS dissection with molecular confirmation and Ag-ELISA, the BLCM estimated the true prevalence and test performance in the absence of a gold standard. The study showed that routine MI in the study abattoirs has low sensitivity, which markedly underestimates the prevalence of BCC when applied alone. The sensitivity of PS dissection and Ag-ELISA was also low, likely reflecting the low infection levels in local cattle breeds. Because full PS dissection is labor-intensive and costly, there is a need for more practical and sensitive diagnostic methods to achieve accurate prevalence estimates of BCC, especially in areas where light infections occur. This study highlights the need to enhance routine MI by considering incisions to the tongue and heart to increase diagnostic sensitivity. It also underscores the importance of improved training and supervision of meat inspectors to boost detection and protect public health. Strengthening national surveillance and reforming meat-inspection practices are therefore recommended for Ethiopian abattoirs.

CRediT authorship contribution statement

Edilu Jorga: Writing – review & editing, Writing – original draft, Investigation, Formal analysis, Data curation. **Inge Van Damme:** Writing – review & editing, Validation, Supervision, Software, Formal analysis, Data curation. **Bizunesh Mideksa:** Writing – review & editing, Supervision, Investigation, Data curation. **Demeke Zewde:** Writing – review & editing, Validation. **Sarah Gabriël:** Writing – review & editing, Validation, Supervision, Resources, Methodology, Formal analysis, Data curation, Conceptualization.

Ethics approval

Not applicable.

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Declaration of competing interest

The authors declare that they have no competing financial and personal relationships with other people or organizations that could inappropriately influence or bias the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.vprsr.2025.101365.

Data availability

The dataset generated and/or analyzed during the current study is available in the zenodo data repository, (https://zenodo.org/doi/10.5281/zenodo.12798335). DNA sequences were deposited in the NCBI database and are publicly available under accession numbers from OM949870 to OM949879.

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